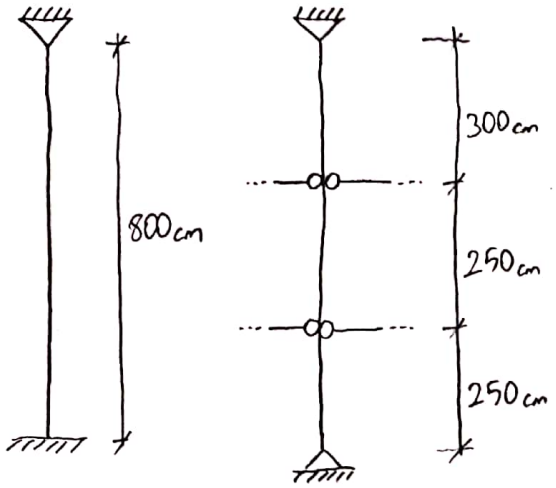


IPN400 profilinden oluşturulmuş, 800 cm uzunluğundaki basınç elemanının kesitten en iyi şekilde istifade edilebilmesi için nasıl yerleştirilmesi gerektiğini bulunuz ve tasarım basınç dayanımını hesaplayınız.

Place the compression member which formed by IPN400 with 800 cm height to utilize its cross-section and compute the design compression strenght.

IPN400

$A = 118\ cm^2$ $d = 400$

$i_x = 15,7\ cm$ $t_w = 14,4$

$i_y = 3,13\ cm$ $h = 323$

$b = 155$ $I_x = 29210\ cm^4$

$t_f = 21,6$ $I_y = 1160\ cm^4$

S355

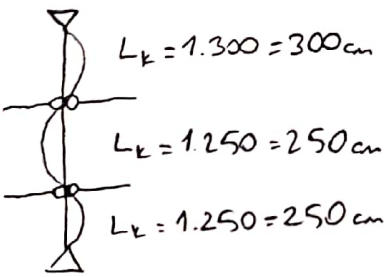
$F_y = 35,5\ kN/cm^2$

$F_u = 51\ kN/cm^2$

$E = 20000\ kN/cm^2$

$$\lambda = \frac{b}{2t_f} = \frac{155}{2.21,6} = 3,59 < \lambda_r = 0,56\sqrt{\frac{E}{F_y}} = 13,29 \rightarrow \text{narin değil (non-slender)}$$

$$\lambda = \frac{h}{t_w} = \frac{323}{14,4} = 22,43 < \lambda_r = 1,49\sqrt{\frac{E}{F_y}} = 35,37 \rightarrow \text{narin değil (non-slender)}$$

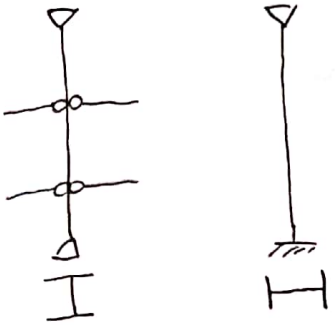




$L_k = 0,7.800 = 560\ cm$

$$\lambda_x = \frac{K_x.L}{i_x} = \frac{L_{kx}}{15,7} = \frac{560}{15,7} = 35,67$$

$$\lambda_y = \frac{K_y.L}{i_y} = \frac{L_{ky}}{3,13} = \frac{300}{3,13} = 95,85$$



CamScanner ile tarandı

$\lambda_r = 4{,}71\sqrt{\frac{E}{F_y}} = 111{,}79$ $\quad \lambda = 95{,}85 < \lambda_r = 111{,}79 \rightarrow$ inelastic buckling (elastik olmayan burkulma)

$F_{cr} = 0{,}658^{F_y/F_e} \cdot F_y$

$F_e = \frac{\pi^2 E}{\lambda^2} = \frac{\pi^2 \cdot 20000}{95{,}85^2} = 21{,}49 \ kN/cm^2$ $\quad F_{cr} = 0{,}658^{35{,}5/21{,}49} \cdot 35{,}5 = 17{,}78 \ kN/cm^2$

$\emptyset P_n = 0{,}9 \cdot 17{,}78 \cdot 118 = 1888{,}32 \ kN$

CamScanner ile tarandı